

# Triangles.

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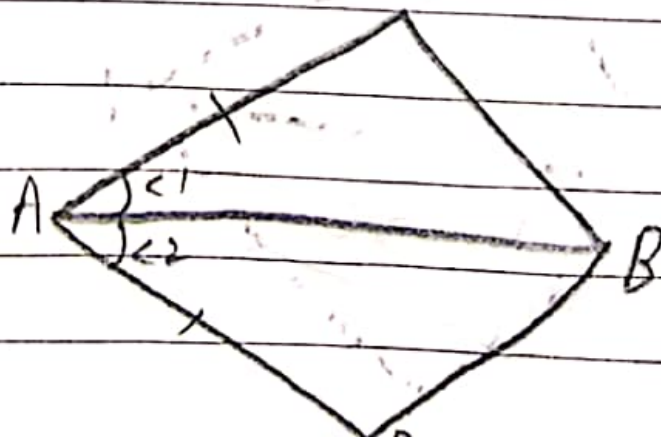
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## ⇒ Congruence of Triangles.

- 1) Rule - 1 = SSS (side side side) rule.
- 2) Rule - 2 = SAS (side Angle side) rule.
- 3) Rule - 3 = ASA (Angle side Angle) rule.
- 4) Rule - 4 = SSA (side side Angle) rule.
- 5) Rule - 5 = AAS (Angle Angle side) rule.
- 6) Rule - 6 = RHS (Right Hypotenuse side) rule.

## EXERCISE - 6.1

Q1) → In Quadrilateral ACBD,  
 $AC = AD$  and  $AB$  bisects  $\angle A$  --- ?

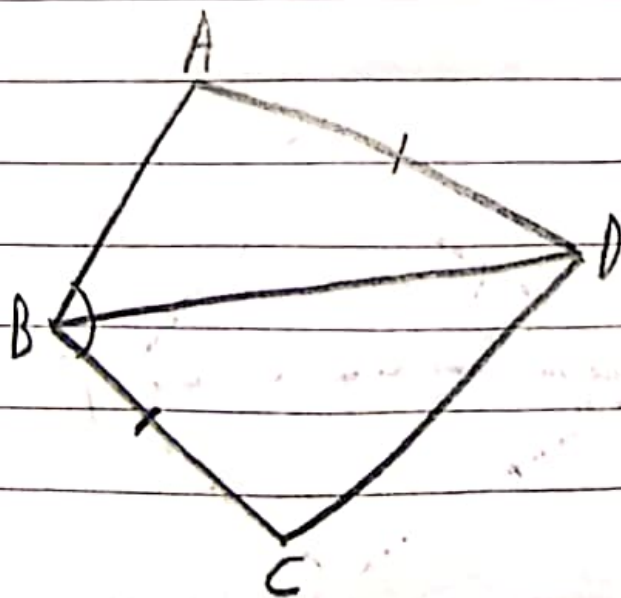


Q1: Given: A Quad. ACBD.  
AB bisects  $\angle A$   
i.e.,  $\angle 1 = \angle 2$   
and  $AC = AD$ .

- To Prove:  $\triangle ABC \cong \triangle ABD$

- Proof: In  $\triangle ABC$  and  $\triangle ABD$   
 $AC = AD$  (Given)  
 $\angle 1 = \angle 2$  (Given)  
 $AB = AB$  (Common)  
- By SAS congruence  
 $\triangle ABC \cong \triangle ABD$   
 $BC = BD$  (C.P.C.T)

Q2)  $ABCD$  is a quadrilateral in which  $AD = BC$   
and  $\angle DAB = \angle CBA$ . Prove that (i)  $\triangle ABD \cong \triangle BAC$   
(ii)  $BD = AC$   
(iii)  $\angle ABD = \angle BAC$ .

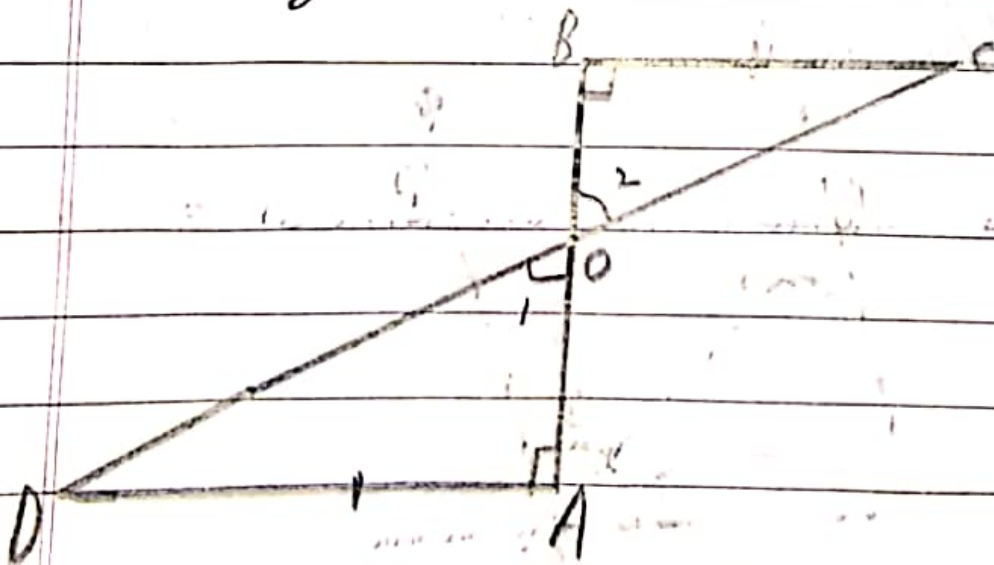


Sol). Given: Quad. ABCD in which,  
 $AD = BC$  and  $\angle DAB = \angle CBA$ .

- To Prove: i)  $\triangle ABD \cong \triangle BAC$   
 ii)  $BD = AC$   
 iii)  $\angle ABD = \angle BAC$

- Proof:  $\angle A = \angle B$  (Given),  
 $AB = AB$  (Common)  
 $BC = AD$  (Given)  
 - By SAS congruence  
 $\triangle ABD \cong \triangle BAC$   
 $BD = AC$  (C.P.C.T)

Q3. AD and BC are equal perpendiculars to a line segment AB. Show that CD bisects AB.



Sol) Given:  $AD \perp AB$ ,  $BC \perp AB$

To Prove:  $CD$  bisects  $AB$  i.e;  $AO = BO$ .

Proof: In  $\triangle BOC$  and  $\triangle DOA$

$$AD = BC \text{ (Given)}$$

$$\angle A = \angle B \text{ (Each } 90^\circ)$$

$$\angle 1 = \angle 2 \text{ (Vertically opposite } \angle\text{'s)}$$

$\therefore \rightarrow$  By AAS Congruence.

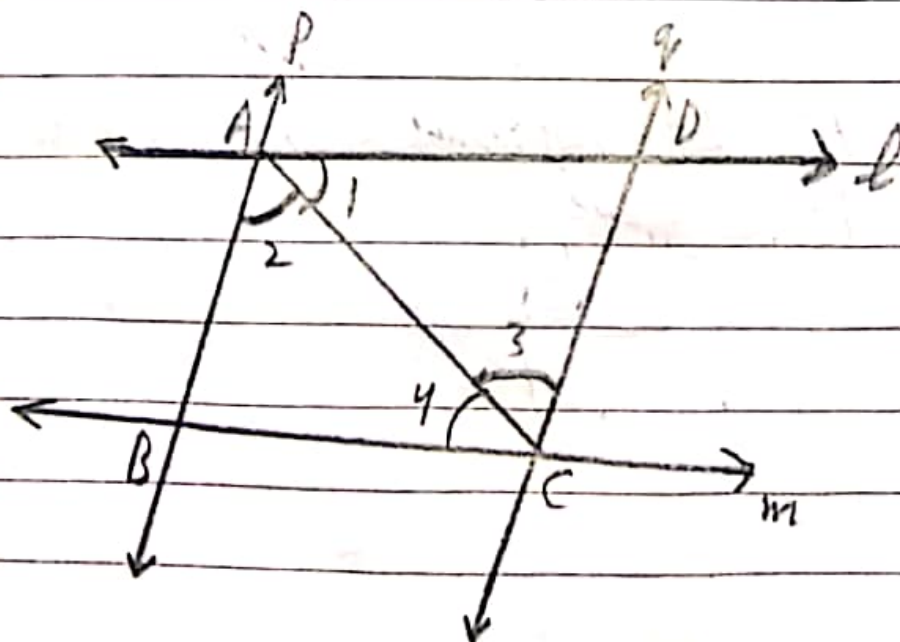
$$\triangle BOC \cong \triangle DOA.$$

$$OA = OB \text{ (C.P.C.T)}$$

$\Rightarrow O$  is the mid point of  $AB$ .

$\Rightarrow CD$  bisects  $AB$  at pt.  $O$ .

Q4)  $l$  and  $m$  are two parallel lines intersected by another pair of parallel lines  $p$  and  $q$ . Show that  $\triangle ABC \cong \triangle CDA$ .



Sol) Given:  $l \parallel m$  and  $p \parallel q$ .

- To Prove:  $\triangle ABC \cong \triangle CDA$ .

- Proof: Since  $l \parallel m$ ,  $AC$  being transversal

$\angle 1 = \angle 4$  (Alt. interior Angles)  
Also,  $p \parallel q$ ,  $AC$  being transversal.

$\angle 2 = \angle 3$  (Alt. interior Angles).

→ Now in  $\triangle ABC$  and ~~ABC~~  $\triangle CDA$

$\angle 4 = \angle 1$  (Proved).

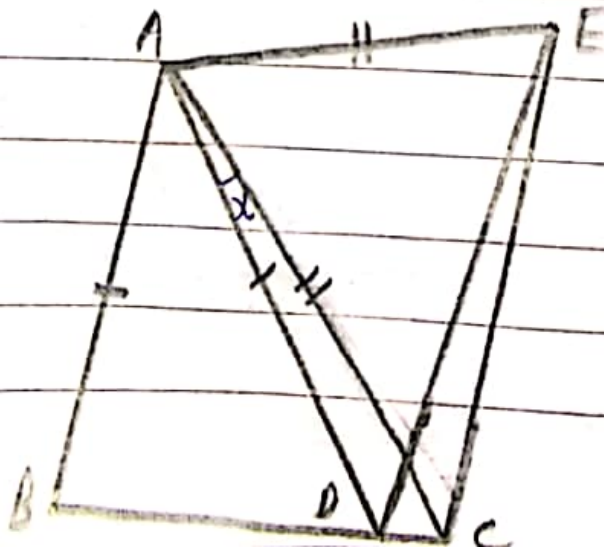
$AC = AC$  (Common)

$\angle 2 = \angle 3$  (Proved).

∴ By ASA Congruence

$\triangle ABC \cong \triangle CDA$ .

Q6:  $AC = AE$ ,  $AB = AD$  and  $\angle BAD = \angle EAC$ . Show that  $BC = DE$ .



Sol) Given:  $AC = AE$ ,  $AB = AD$  and  $\angle BAD = \angle EAC$

- To Prove:  $BC = DE$ .

- Proof: Since  $\angle BAD = \angle EAC$

- Adding angle  $x$  both sides, we have

$$\angle BAD + x = \angle EAC + x$$

$$\Rightarrow \angle BAC = \angle EAD \text{ ---- } \textcircled{1}$$

- Now, In  $\triangle ABC$  and  $\triangle ADE$

$$AC = AE \text{ (Given)}$$

$$\angle BAC = \angle EAD \text{ (Proved above).}$$

$$AB = AD \text{ (Given)}$$

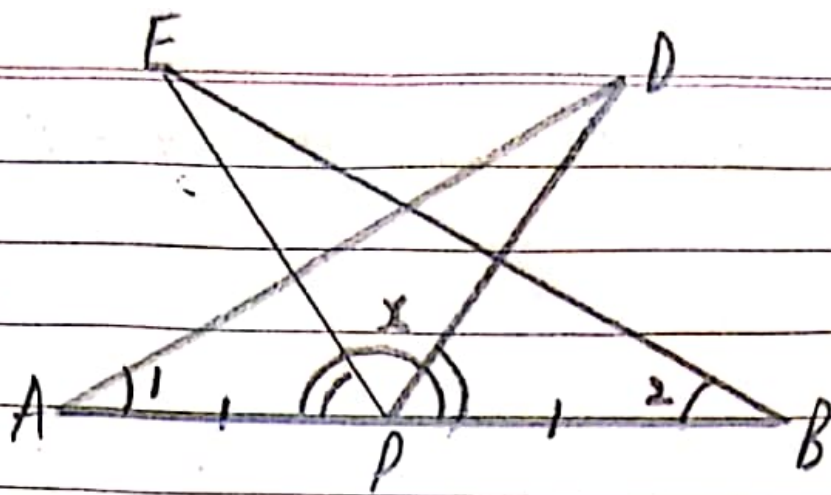
$\therefore$  By SAS rule.

$$\triangle ABC \cong \triangle ADE$$

$$\Rightarrow BC = DE \text{ (by C.P.C.T.)}$$

Q71 AB is a line segment and P is the mid-point. D and E are points on the same side of AB such that  $\angle BAD = \angle ABE$  and  $\angle EPA = \angle DPB$ . Show that: (i)  $\triangle DAP \cong \triangle EBP$ .

$$(ii) AD = BE.$$



Sol) Given:- P is the mid-point of AB i.e;  $AP = BP$ .  
 $\angle BAD = \angle ABE$  and  $\angle EPA = \angle DPB$ .

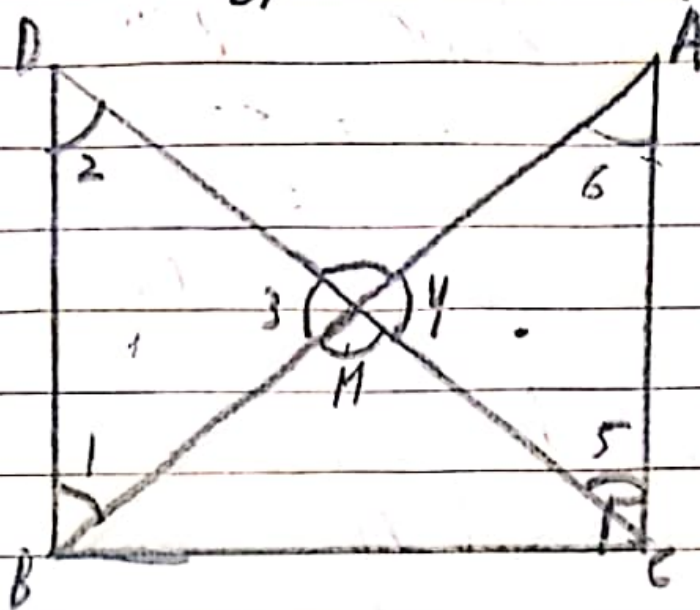
- To Prove:- (i)  $\triangle DAP \cong \triangle EBP$ .  
 (ii)  $AD = BE$ .

- Proof:- Since  $\angle EPA = \angle DPB$ .  
 Adding angle x both sides.  
 $\angle EPA + x = \angle DPB + x$   
 $\Rightarrow \angle APD = \angle BPE$  ---- ①

- Now in  $\triangle DAP$  and  $\triangle EBP$ .  
 $\angle 1 = \angle 2$  (Given).  
 ~~$\angle APD = \angle BPE$~~   
 $\angle APD = \angle BPE$  (Proved).  
 $AP = BP$  (Given)  
 $\therefore$  By ASA Congruence  
 $\triangle DAP \cong \triangle EBP$   
 $AD = BE$  (C.P.C.T).

Q0) In right  $\triangle ABC$ , right angled at  $C$ ,  $M$  is the mid-point of hypotenuse  $AB$ ....?

Sol)



- Given:  $ABC$  is a right angled  $\triangle$ , right Angled at  $C$ .
- $M$  is the mid point at  $AB$  i.e;  $AM = MB$ , Also  $DM = CM$
- To Prove:-
  - 1)  $\triangle AMC \cong \triangle BMD$
  - 2)  $\angle DBC$  is a rt. angle.
  - 3)  $\triangle DBC \cong \triangle ACB$
  - 4)  $CM = \frac{1}{2} AB$ .

(i)- Proof 1:- In  $\triangle AMC$  and  $\triangle BMD$   
 $AM = BM$  (Given)  
 $\angle 3 = \angle 4$  (Vertically Opp. Angle)  
 $CM = DM$  (Given)  
 $\therefore$  By SAS Congruence.

⇒

$$\triangle AMC \cong \triangle BMD$$

(ii) Since  $\triangle AMC \cong \triangle BMD$

$$\angle 2 = \angle 5$$

$$\angle 1 = \angle 6$$

$$DB = AC \text{ (C.P.C.T.)}$$

- Taking line segments AC and DB, with transversal AB.

$$\angle 1 = \angle 6 \text{ (Proved).}$$

- Alternate angles are equal,  
⇒  $AC \parallel DB$ .

- Now  $AC \parallel DB$ , BC being Transversal

$$\Rightarrow \angle B = \angle C,$$

$$\text{But } \angle C = 90^\circ$$

$$\Rightarrow \angle B = 90^\circ$$

$$\Rightarrow \angle DBC \text{ is a rt. Angle}$$

(iii) In  $\triangle DBC$  and  $\triangle ACB$

$$DB = AC \text{ (Proved)}$$

$$\angle B = \angle C \text{ (} 90^\circ \text{)}$$

$$BC = BC \text{ (Common)}$$

- By SAC Congruence,  
 $\triangle DBC \cong \triangle ACB.$

(iv) Since  $\triangle DBC \cong \triangle ACB$

- $CD = AB$
- $CM + MD = AB$  ( $\because M$  is the mid-point).
- $CM + CM = AB$  ( $MD = CM$ )
- $2CM = AB$
- $CM = \frac{AB}{2}$

$$\Rightarrow \boxed{CM = \frac{1}{2} AB.}$$